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27 May 1960
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27 March 1958

MILITARY
SPECIFICATION

REPEATER, TELEGRAPH TH-18( )/FG

1. SCOPE

1.1 This specification covers one type of telegraph regenerative repeater, capable of operating in any one way d.c. neutral telegraph loop of 20 or 60 milliamperes at speeds of 60, 75 and 100 words per minute, designated as Repeater, Telegraph TH-18( )/FG. (See 6.3)

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids, form a part of this specification:

SPECIFICATIONS

FEDERAL

UU-T-111	Tape; Paper, Gummed (Sealing and Securing)
PPP-P-291	Paperboard, Wrapping, Cushioning.
PPP-B-566	Boxes, Folding, Paperboard.
PPP-B-585	Boxes, Wood, Wirebound.
PPP-B-591	Boxes, Fiberboard, Wood Cleated.
PPP-B-601	Boxes, Wood, Cleated Plywood.
PPP-B-621	Boxes, Wood, Nailed and Lock-Corner.
PPP-B-636	Boxes, Fiber.
PPP-C-843	Cushioning Material, Cellulosic.

MILITARY

MIL-E-75	Packaging, Packing and Container-Marking of Electron Tubes.
MIL-P-116	Preservation, Methods of
MIL-B-117	Bags, Interior Packaging.
MIL-T-152	Treatment, Moisture-and-Fungus-Resistant, of Communications, Electronic, and Associated Electrical Equipment.
MIL-B-10377	Box, Wood, Cleated, Veneer, Paper-Overlaid.

FSC 5805

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MIL-I-11748	Interference Reduction for Electrical and Electronic Equipment.
MIL-M-13231	Marking of Electronic Items
MIL-F-14072	Finishes for Ground Signal Equipment.
MIL-E-15090	Enamel, Equipment, Light-Gray (Formula No. 111)

STANDARDS

MILITARY

MIL-STD-105	Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-129	Marking for Shipment and Storage
MIL-STD-169	Extreme-Temperature Cycle.
MIL-STD-170	Moisture Resistance Test Cycle for Ground Signal Equipment.
MIL-STD-252	Wired Equipment, Classification of Visual and Mechanical Defects.

DRAWINGS

SIGNAL CORPS

SC-C-33073	Form to be Used for Pilot Pack Process Sheet
SC-DL-349080	Repeater, Telegraph TH-18( )/FG.

(Copies of specification, standards, and drawings required by contractors in connection with specific procurement functions should be obtained from the procuring agency or as directed by the contracting officer. Both the title and identifying number or symbol should be stipulated when requesting copies.)

2.2 Other publications.- The following documents form a part of this specification. Unless otherwise indicated, the issue in effect on date of invitation for bids shall apply.

AMERICAN TRUCKING ASSOCIATION, INC.

National Motor Freight Classification

(Application for copies should be addressed to the National Classification Board, 1424, 16th Street, N.W., Washington 6, D.C.)

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CONSOLIDATED CLASSIFICATION COMMITTEE

Consolidated Freight Classification
Uniform Freight Classification

(Applications for copies should be addressed to Consolidated Classification Committee, 202 Union Station, Chicago 6, Illinois.)

3. REQUIREMENTS

3.1 Description.- Repeater, Telegraph TH-18( )/FG consists of a drawer type chassis with a front panel. Four plug in assemblies and two power supplies contain the tubes, transformers, and other major components. The main chassis assembly contains the front panel controls and indicators, a terminal board at the rear (for external connections) and the interconnecting wiring. Two power supplies are included in the repeater, one for the repeater tube circuits and one for the output loop supply. Both supplies are of the regulated type, using series regulator tubes. The chassis for each power supply may be removed but they are not plug-in units. Electrical connections from the power supplies are wired directly to the main chassis.

3.2 Preproduction samples.- The contractor shall furnish preproduction samples for approval, if required by the invitation for bids and contract. (See 4.3 and 6.2(d)(1).)

3.3 Construction.- Repeater, Telegraph TH-18( )/FG shall be constructed in accordance with Drawing SC-DL-349080.

3.4 Use of virgin materials in plastic and rubber compounds for insulation and jacket of wires and cables.- Only virgin materials shall be used for plastic and rubber compounds intended for insulation and jacket of wires and cables.

3.5 Marking.-

3.5.1 General.- Marking shall conform to Specification MIL-M-13231.

3.5.2 Visibility.- Wherever practicable, parts shall be so mounted that their identification marking will be readily visible with minimum disassembly of the equipment.

3.6 Serial numbers.- Repeater, Telegraph TH-18( )/FG shall be serial numbered.

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3.7 Finish, protective.- The equipment shall be given protective finish in accordance with Specification MIL-F-14072. The final paint film on type 1 surfaces shall be semigloss light-gray enamel conforming to Specification MIL-E-15090.

3.8 Electrical requirements.-

3.8.1 Frequency coverage and regenerative capabilities.- The equipment shall be capable of accepting 7.42 unit start-stop teletype signals at speeds of 60, 75 or 100 words per minute with up to 45 percent peak distortion at each speed, and shall be capable of reforming and retiming these signals so that the output shall have a maximum of 2 percent peak distortion.

3.8.2 Loop current.-

3.8.2.1 Input.- The receive side of the repeater shall be capable of accepting either 20 ma or 60 ma neutral signals.

3.8.2.2 Output.- The transmit side of the repeater shall be capable of supplying 20 ma or 60 ma direct current for neutral operation. An internal rheostat shall be provided to allow operation into circuits from 0 to 3,500 ohms.

3.8.3 Short start pulse rejection.- The repeater shall not be activated by spacing pulses whose time duration is less than 1/2 of a baud.

3.8.4 Maximum current handling capacity.- The input and output circuits shall be capable of withstanding 100 ma current.

3.8.5 Distortion.- The repeater shall accurately regenerate incomming signals having transition time displacements up to 45 percent of a unit baud with less than 2 percent distortion in outputs.

3.8.6 Simultaneous output operation.- The two output ciruits shall work separately or together without interference to either circuit.

3.8.7 Power supply.- The repeater shall be capable of operating from a power source of 115/230 volts ± 10 percent, 50 to 60 cycles ± 3 cycles.

3.8.8 Suppression of radio interference.- The equipment shall meet the emanation and susceptibility limits of Specification MIL-I-11748 for class 1. (See 4.4) Application of suppression components and techniques shall conform to that specification. If the contractor considers it desirable to modify the equipment proper in order to facilitate conformance with the specified emanation and susceptibility limits, he shall obtain approval from the contracting officer before proceeding with the modification.

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3.9 Service conditions.- The equipment shall meet the following service conditions:

3.9.1 Operation.- Continuous use for a period of 1 year, under the worldwide environmental conditions specified in 3.9.2 through 3.10, with no more than normal maintenance and replacement of parts. Where the contractor is required to make a selection of parts, materials, processes, construction methods, etc, he shall be guided by this requirement. Approval of the preproduction sample (see 3.2) shall be considered as compliance with this requirement.

3.9.2 Temperature (see 4.9.1).-

(a) Operating: Ambient temperature in the range of +125°F to -25°F. (The 125°F temperature includes effect of sun-load.) Exposure at the high temperature extreme not to exceed 4 hours, and at the low temperature extreme not to exceed 72 hours, at any one time.

(b) Nonoperating: Exposure in the range of +160°F to -80°F; exposure at the high temperature extreme not to exceed 4 hours, and at the low temperature extreme not to exceed 24 hours, at any one time.

3.9.3 Relative humidity (see 4.9.2).- Up to 97 percent relative humidity for 20 hours; and exposure at 100 percent relative humidity, with condensation, for 4 hours.

3.9.4 Elevation (see 4.9.3).-

(a) Operating: Up to 10,000 feet above sea level.

(b) Nonoperating: Up to 50,000 feet above sea level.

3.10 Vibration and shock.-

3.10.1 Vibration, internal.- With the equipment shock-mounts blocked, the amplitude of vibration of any part, subassembly, or structural member of the equipment shall not exceed twice the amplitude of the vibration applied to the equipment at any frequency between 10 and 55 cycles per second. (See 4.9.4)

3.10.2 Shock, benchhandling.- The equipment shall meet full specification performance and show no physical damage when subjected to the test of 4.9.5.

3.11 Preconditioning.- The equipment shall be capable of meeting the requirements herein, without subsequent processing, after subjection to the bounce preconditioning of 4.6. (Also see 4.5)

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3.12 Interchangeability.- Like units, assemblies, subassemblies, and replaceable parts shall be physically and functionally interchange, without modification of such items or of the equipment. (See 4.11) Individual items shall not be hand picked for fit or performance. Reliance shall not be placed on any unspecified dimension, rating, characteristic, etc.

3.13 Technical literature and running spare parts.- Technical literature and running spare parts shall be furnished as specified by the contract. Running spare parts shall be identical to corresponding parts in the equipment furnished on the order.

3.14 Workmanship.- The equipment shall be manufactured and assembled in accordance with the applicable portions of the following paragraphs herein.

3.3 Construction
3.4 Use of virgin materials in plastic and rubber compounds for insulation and jacket of wires and cables.
3.5 Marking
3.7 Finish protective.

4. QUALITY ASSURANCE PROVISIONS

4.1 Contractor responsibility.- Unless otherwise specified herein the supplier is responsible for the performance of all inspection requirements prior to submission for Government inspection and acceptance. Except as otherwise specified, the supplier may utilize his own facilities or any commercial laboratory acceptable to the Government. Inspection records of the examinations and tests shall be kept complete and available to the Government as specified in the contract or order.

4.2 Classification of inspection.- Inspection shall be classified as follows:

(a) Preproduction inspection (does not include preparation for delivery) (See 4.3)

(b) Acceptance inspection of equipment before preparation for delivery. (See 4.4 and 4.5)

(1) Acceptance inspectance of preparation for delivery. (See 4.12)

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4.3 Preproduction inspection.- This inspection will be performed by the Government unless otherwise specified in the contract. It shall consist of the preproduction inspection specified in table I, the inspection specified in the subsidiary documents covering the items listed in 4.4, and the inspection specified for Group A, group B, and group C (See tables II, III and IV respectively). The preproduction inspection will normally be performed in this order: (1) vibration, (2) shock, benchhandling; other preproduction inspection may precede, follow, or be interspersed between the foregoing.

Table I - Preproduction Inspection

Inspection	Rqt. Par.	Insp. Par.
Suppression of radio interference	3.8.8	4.8.6
Temperature	3.9.2	4.9.1
Moisture resistance	3.9.3	4.9.2
Elevation	3.9.4	4.9.3
Vibration	3.10.1	4.9.4
Shock; bench handling	3.10.2	4.9.5

4.4 Acceptance inspection covered by subsidiary documents.- The following shall be inspected under the applicable subsidiary documents as part of the acceptance inspection before preparation for delivery:

Item	Where required
Marking	3.5
Finish	3.7

4.5 Acceptance inspection of equipment before preparation for delivery.- The contractor, to demonstrate compliance with specified requirements, shall perform the inspection specified in 4.4 and 4.5.1 through 4.5.5. This does not relieve the contractor of his responsibility for performing any additional inspection which is necessary to control the quality of the product and to assure compliance with all specification requirements. The Government will review and evaluate the contractor's inspection procedures and examine the contractor's inspection records. In addition the Government--At its discretion--may perform all or any part of the specified inspection, to verify the contractor's compliance with specified requirements. Each unit which will be subjected to group A, group B, or group C inspection shall be preconditioned after final assembly. (See 3.11)

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4.5.1 Group A inspection.- This inspection (including sampling) shall conform to Table II and Standard MIL-STD-105. Unless otherwise specified, normal inspection shall be used at the start of the contract. Group A inspection shall be performed in any order which is satisfactory to the Government.

Table II - Group A Inspection

Inspection	Requ. Par.	Insp. Par.	AQL Major	AQL Minor
Visual and Mechanical	3.14	4.10		
Input chassis			1.0 dphu	4.0 dphu
Timing chassis			1.5 dphu	6.5 dphu
Sampling chassis			1.5 dphu	6.5 dphu
Output circuits chassis			0.65 dphu	1.5 dphu
Main power supply			0.65 dphu	1.5 dphu
Loop power supply			0.65 dphu	1.5 dphu
Overall repeater TH-18( )/FG			1.0 dphu	4.0 dphu
Electrical				
Loop current				
Input	3.8.2.1	4.8.1		
Output	3.8.2.2	4.8.1	1.0 dphu	*
Short pulse rejection	3.8.3	4.8.2		
Distortion	3.8.5	4.8.4		
Operation				
60 WPM input only				
Nominal input power (115V, 60 cycle)	3.8.7	4.7	1.0%	*
Simultaneous output	3.8.6	4.8.4		

* All electrical and operational defects are of major category.

4.5.2 Group B inspection.- This inspection, including sampling shall conform to table III and to the special procedures for small-sample inspection of Standard MIL-STD-105. The reduced inspection procedure shall be R-1, Group B inspection shall normally be performed on inspection lots that have passed group A inspection and on samples selected from units that have been subjected to and met the group A inspection. The AQL shall be 4.0 percent defective and the inspection level shall be L-8 for normal and tightened, and L-6 for reduced.

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4.5.2.1 Order of inspection within group B.- Group B inspection shall be performed in any order which is satisfactory to the Government.

4.5.2.2 Procedure in case of failure.- When an inspection lot fails, the contractor shall immediately investigate the cause of failure and take corrective action to assure that subsequent lots do not contain the same defect or defects.

Table III - Group B Inspection

Inspection	Reqt. Par.	Insp. Par.	AQL
Maximum current handling capacity	3.8.4	4.8.3	4
Operation			
75 WPM and 100 WPM input signal	3.8.1	4.7	4
Power input - 50/60 cps ± 3 cps, 115/230 volts ± 10.0%	3.8.7	4.8.5	4

4.5.3 Group C inspection.- This inspection shall be as listed in table IV, and shall normally be performed on sample units that have been subjected to and met group A and group B inspection.

Table IV - Group C Inspection

Inspection	Reqt. Par.	Insp. Par.
Interchangeability	3.12	4.11

4.5.3.1 Sampling for group C inspection of equipment.- Two repeaters for each 50 produced (on contract or purchase order) shall be selected without regard for their quality.

4.5.3.2 Noncompliance.- If a sample unit fails group C inspection, the contractor shall immediately investigate the cause of failure and shall report to the Government inspector the results thereof and details of the corrective action taken on the process and all units of product which were manufactured with the same conditions, materials, processes, etc. If the Government inspector does not consider that the corrective action will enable the product to meet specified requirements, or if the contractor cannot determine the cause of failure, the matter shall be referred to the contracting officer. (See 6.5)

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4.5.4 Reinspection of conforming group B and group C sample units.- Unless otherwise specified, sample units which have been subjected to and passed group B or group C inspection, or both, may be accepted on contract, provided that they are resubjected to and pass group A inspection after repair of all visible damage.

4.5.5 Disposition of nonconforming product.- Disposition of nonconforming product shall conform to Standard MIL-STD-105. Such product shall be suitably tagged or identified to indicate the cause of failure and means employed to correct the fault. The required information shall be presented to the Government when the product is submitted.

4.6 Bounce preconditioning.- The unit, with shock mounts removed or blocked, shall be placed in its normal operation position on the table of the Package Tester as made by the L.A.B Corporation, Skaneateles, New York, or equal. The package tester, shafts in phase shall have a speed such that it is just possible to insert a 1/32 inch thick strip of material under one corner or edge of the unit to a distance of 3 inches as the unit bounces. The unit shall be subjected to this preconditioning for 1 minute. After bounce preconditioning, the unit shall not be repaired, alined, cleaned, or otherwise changed prior to subjection to acceptance inspection.

4.7 Operational.- The telegraph repeater shall be alternately adjusted for the specified mode of operation in tables II and III (See 3.8.1). Satisfactory retransmission of the input signal, in conformance with the requirements herein, in each combination of service power and frequency specified (see tables II and III, shall be an indication of acceptable operational performance.

4.8 Electrical test.- The test set up for all electrical tests shall be as shown in Figure 1 unless otherwise specified.

4.8.1 Loop current.- After a 10 minute warm-up of the equipment input signals shall be fed into the repeater from Test Set TS-2( )/TG. The input shall be adjusted to 60 ma and the output loop shall be adjusted to 60 ma. The output teletypewriter shall reproduce input signals at 60 wpm. This test shall be repeated at 20 ma.

4.8.2 Short start rejection.- The input signals shall be adjusted to 61 percent marking distortion. Signals transmitted with this adjustment shall fail to strike any characters on the teletypewriter.

4.8.3 Current handling capacity.- The input and output current shall be increased to 100 ma with the 60 ma input switch closed. Components shall not be damaged by this overload.

4.8.4 Distortion.- The input signal shall be set to 45 percent distortion. A teletypewriter and distortion analyzer shall be connected to both outputs. The distortion in either output shall be less than 5 percent. Removing either output shall not affect the other.

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4.8.5 Power Regulation.- The input voltage source to the repeater shall be varied plus and minus 10 percent at 115V and 230V, and plus and minus 3 percent at 50 and 60 cycles per second and the equipment shall pass the operational test of 4.7.

4.8.6 Suppression of radio interference.- The equipment shall be tested in accordan with MIL-I-11748 to determine compliance with 3.8.8.

4.9 Service conditions test.-

4.9.1 Temperature test.- The equipment shall be subjected to the temperature cycle shown on MIL-STD-169. Tests shall be made at step 4 and step 9 in addition to the steps required in MIL-STD-169. The equipment shall meet the requirements of 3.8.

4.9.2 Moisture-resistance test.-

4.9.2.1 Test conditions.-

(a) Do not remove equipment from the humidity chamber for measurements.

(b) Start measurements not more than 5 minutes after power is applied to the equipment. Complete measurements as rapidly as possible. Do not leave power on after measurements have been completed.

4.9.2.2 Test procedure.- The sample equipments will be tested as follows:

(a) Dry at 130° ± 5° F for 24 hours.

(b) Condition at 77° ± 5° F and 40 to 50 percent relative humidity for 24 hours.

(c) Measure the performance and readjust if necessary to meet specification requirements.

(d) Subject to continuous cycling for a total of five 48-hour cycles. Temperature, relative humidity, and period of time for each portion of the cycle shall conform to MIL-STD-170. Measure performance in accordance with the periods indicated on drawing for the first four cycles.

(e) After cycling has been completed, condition the equipments for 24 hours at 77° ± 5° F at 40 to 60 percent relative humidity. No repair or replacement of parts shall be made. After adjustment, the equipments shall meet the performance specified in 4.9.2.3.

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4.9.2.3 Performance.- Repeat the tests of 4.9.2.2. There shall be no mechanical deterioration, as evidenced by corrosion of metal parts of binding of rotating parts, and there shall be no degradation in performance from that obtained prior to the five humidity cycles.

4.9.2.4 Failure.- If any component fails to meet the performance specified in 4.9.2.3 during cycling, it shall be realined or readjusted once. If the component then fails to meet 4.9.2.3 or fails subsequently during cycling, it does not pass the test. In addition, if the component fails to meet full specification requirements after conditioning and adjustment, as specified in 4.9.2.2(e), it does not pass the test.

4.9.3 Elevation test.- The equipment, placed in its normal operating position in an altitude chamber, shall be tested as follows: (Note: To check temperature stabilization, place a thermocouple on the largest internal mass centrally located within the equipment.)

(a) With the equipment placed in the chamber, operating and stabilized at room temperature, the equipment shall be tested in accordance with 4.8.

(b) Reduce the barometric pressure to 20.6 ± 0.1 inches of mercury (simulated 10,000 feet above sea level) and, after temperature and pressure stabilization, repeat the measurements specified in 4.9.3(a). The equipment shall meet specified performance for those measurements.

(c) Remove the power from the equipment and further reduce the barometric pressure to 3.4 ± 0.1 inches of mercury (simulated 50,000 feet above sea level). Maintain this condition for 1 hour.

(d) Return the chamber to ambient pressure. Apply power to the equipment and, after stabilization, repeat the measurements specified in 4.9.3(a). The equipment shall meet specified performance for these measurements.

4.9.4 Test for internal vibration.- Internal vibration of the equipment shall be measured as follows, to determine conformance to 3.10.1:

(a) Secure the equipment (shock mounts, removed or blocked) directly to a vibration table that can be controlled within 10 percent of the specified amplitude. Mounting method shall be such that vibration within the equipment can be observed and measured. To facilitate this observation and measurement, subassemblies may be tested separately provided they are secured to the table in a manner similar to that used to mount them in the equipment.

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(b) Vibrate the equipment successively in three mutually perpendicular directions over a frequency range of 10 to 55 cycles per second. The total excursion of the applied vibration shall be not less than 0.030 inch.

(c) In each of the three directions, change the frequency in steps of one cycle per second and maintain each frequency for at least 10 seconds.

(d) Measure vibration amplitudes by optical means, or by other means provided that vibration of the part is not affected by the measurement.

4.9.5 Shock test: bench-handling.- The chassis-and-front-panel assembly shall be removed from its enclosure, as for servicing, and placed in a suitable position for servicing on a solid, 2-inch fir bench top. The test shall be performed as follows, in a manner simulating shocks liable to occur during servicing:

(a) Tilt up the assembly through an angle of 30 degrees, using one edge of the assembly as an axis, and permit the assembly to drop back freely to the horizontal. Repeat, using other practicable edges of the same horizontal face as axes, for a total of four drops.

(b) Repeat "(a)", with the assembly resting on other faces, until it has been dropped for a total of four times on each face on which the assembly might practically be placed during servicing.

4.10 Visual and mechanical inspection.- The equipment shall be examined for the defects listed in Standard MIL-STD-252.

4.11 Inspection for interchangeability.- The dimensions listed below shall be gaged or measured to determine compliance with the physical interchangeability requirements of 3.12. When a listed dimension is not within specified or design limits, it shall be considered a major defect.

(a) External and internal dimensions of insertable subassemblies, when such dimensions affect mating of parts.

(b) Internal dimensions of cavities in assemblies, when such dimensions affect acceptability of insertable subassemblies.

(c) Location of connectors, locking pins, fasteners, slides, and mountings which receive mating parts of plug-in assemblies and major components.

(d) Location of connectors, locking pins, fasteners, slides, and mountings on plug-in assemblies.

(e) Size and form of special threads.

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4.12 Acceptance inspection of preparation for delivery.- Preparation for delivery shall be inspected in accordance with Specification MIL-P-116 to determine conformance to the requirements of section 5.

4.13 Rough handling test (preparation for delivery).- When rough handling test in accordance with Specification MIL-P-116 is required by the contract (see 6.2(d)(2)c, the following functional tests shall be conducted to determine freedom from operational malfunction caused by the rough handling.

4.7 Operational test.

5. PREPARATION FOR DELIVERY

5.1 Preservation and packaging.-

5.1.1 Level A.- Repeater, Telegraph, TH-18( )/FG shall be preserved, packaged, and tested in accordance with the procedures specified for the designated methods as prescribed in Specification MIL-P-116 and as described herein.

5.1.1.1 Technical manual.- Each technical manual shall be packaged Method IC-6 as follows: Inclose each technical manual within a type II, class d bag conforming to Specification MIL-B-117. Heat seal the bag to effect closure.

5.1.1.2 Tubes (spare).- The spare tubes shall be packaged individually following the applicable procedures specified in Specification MIL-E-75.

5.1.1.3 Fuses (spare).- Spare fuses shall be packaged Method III as follows: Fuses of like stock number shall be grouped together and each group cushioned by wrapping in type I, wrapping cushioning paperboard conforming to Specification PPP-P-291. Secure the cushioning with class I gummed paper tape conforming to Specification UU-T-111. Place each cushioned group of fuses within a close-fitting style II, Type A, class (a) paperboard box conforming to Specification PPP-B-566. Box closure shall be in accordance with the requirements of the box specification.

5.1.1.4 Relay, polar.- (spare) Each relay shall be packaged Method IC-6 as follows: Cushion each relay by wrapping within a single thickness of cellulosic cushioning material, (1/4 inch thick), type II conforming to Specification PPP-C-843. Place the cushioned relay within a type II, class d bag conforming to Specification MIL-B-117. Heat seal the bag to effect closure. Place the bagged relay within a close-fitting paperboard box. Box closure shall be in conformance with Specification PPP-B-566.

5.1.1.5 Repeater, Telegraph TH-18( )/FG.-

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5.1.1.5.1 Unit packaging.- Each repeater shall be packaged Method IC-5 as follows: All components within the chassis shall be blocked and braced with cells or pads or both fabricated of double faced corrugated fiberboard class 2, Specification PPP-B-636, and designed to prevent independent movement of the individual components. The repeater shall then be cushioned on all surfaces with cells or pads or both, fabricated of double faced corrugated fiberboard, and designed to absorb the shock of impact encountered in handling and transit. Cells and pads shall have water-resistant qualities equal to the box for which they are intended. Place the cushioned repeater within a close-fitting, regular slotted style fiber box, W5c, class 2 conforming to Specification PPP-B-636. Place the technical manual (see 5.1.1.1) between the contents and upper flaps of the box. Box closure shall be in accordance with the referenced box specification.

5.1.2 Level C.- Repeater, Telegraph TH-18( )/FG shall be preserved and packaged in accordance with commercial practice and in a manner that will afford protection against corrosion, deterioration and physical damage during direct shipment to the first receiving activity.

5.2 Packing.-

5.2.1 Level A.- Repeater, Telegraph TH-18( )/FG packaged as specified above together with the consolidated packaged shall be packed in shipping containers conforming to the requirements of the specifications referenced below for the designated level. The gross weight shall not exceed the weight shown. Closure shall be as prescribed in the applicable box specifications or appendix thereto.

	Box Specification	Max. Gross Wt.
Fiberboard(Type I, Class 2)	PPP-B-636	Table II
Nailed Wood(Style 4,Class 2)	PPP-B-621	200 lbs.
Wirebound(Style 2 or 3,Class 2)	PPP-B-[illegible]	200 lbs.
Wood Cleated Fiberboard(Overseas)	PPP-B-591	200 lbs.
Cleated Plywood (Overseas)	PPP-B-601	200 lbs.
Wood Cleated, Veneer, Paper Overlaid (Overseas)	MIL-B-10377	200 lbs.
Nailed Wood (Style 2, Class 2)	PPP-B-621	Specification Limit

5.2.1.1 Metal strapping.- Shipping containers shall be strapped in conformance with the requirements of the appendix of the applicable container specification only for direct shipment to ports.

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5.2.2 Level B.-

	Box Specification	Max. Gross Wt.
Fiberboard(Type 1, Class 1)	PPP-B-636	Table I
Nailed Wood(Style 4, Class 1)	PPP-B-621	200 lbs.
Wirebound(Style 3, Class 1)	PPP-B-585	200 lbs.
Wood Cleated Fiberboard(Domestic)	PPP-B-591	200 lbs.
Cleated Plywood (Domestic)	PPP-B-601	200 lbs.
Wood Cleated,Veneer, Paper Overlaid (Domestic)	MIL-B-10377	200 lbs.
Nailed Wood(Style 2, Class 2)	PPP-B-621	Specification Limit

5.2.3 Level C.- Repeater, Telegraph TH-18( )/FG shall be packed for shipment in a manner conforming to the requirements of Uniform Freight Classification No. 4 and Consolidated Freight Classification, Rule 21 for rail shipment, National Motor Freight Classification No. A-3 and No. 14 for truck shipment, Parcel Post Regulations and the regulations of other carries as applicable to the mode of transportation employed at lowest transportation rate.

5.3 Marking.- Interior packages and exterior shipping containers shall be marked in accordance with the applicable provisions of Standard MIL-STD-129.

6. NOTES

6.1 Intended use.- Repeater, Telegraph TH-18( )/FG is a telegraph regenerative repeater used to correct telegraph signal distortion.

6.2 Ordering data.- Procurement documents should specify the following:

(a) Title, number, and date of this specification and any amendment thereto.

(b) Level of packaging and level of packing required for shipment. (Level A, Level B, or Level C.)

(c) The specific paragraphs of section 5 which are applicable to the particular procurement.

(d) Preproduction inspection.

(1) Two samples of each item cited in section 1 are generally required so that lengthy environmental tests can be completed on one sample while complete performance measurements can be made on the second sample.

(2) Preproduction pack(s) as follows:

a. Makeup of pack(s).
b. Number of each kind of pack to be submitted.
c. Inspection to be performed thereon--including rough handling test, which will not be performed as procurement inspection.

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(e) Marking and shipping of samples.

(f) Place of final inspection.

(g) Technical literature required.

(h) Quantity of tools and running spare parts required.

6.3 Nomenclature.- The parentheses in the nomenclature will be deleted or replaced by a letter identifying the particular design; for example: TH-18W/FG. The contractor should apply for nomenclature in accordance with the applicable clause in the contract.

6.4 Verification inspection.- The amount of verification inspection (by the Government) will be adjusted to make maximum utilization of the contractor's quality control system and the quality history of the product, and will normally be identified by the categories listed below:

(a) Type A--The total of that inspection set forth in the Quality Assurance Provisions of this specification or the contract. Included in this category is that amount of inspection referred to as normal and tightened inspection by Militar Standard 105.

(b) Type B--That inspection set forth in the Quality Assurance Provisions of this specification or the contract reduced in amount under the reduced inspection provisions of Military Standard 105.

(c) Type C--A reduced inspection procedure resulting in a material reduction in the amount of inspection set forth in the Quality Assurance Provisions of this specification. The amount of inspection is less than that provided for in Type B and is based upon a consistently acceptable product resulting from a planned quality control system voluntarily employed by the contractor in the production of the product.

6.5 Group C inspection.- Approval to ship may be withheld, at the discretion of the Government, pending the decision from the contracting officer on the adequacy of corrective action. (See 4.5.3.2)

6.6 Definition of level B.- Level B packaging requirements detailed herein are designed to furnish protection against corrosion, deterioration, and physical damage during domestic shipments; covered storage for a period of three years or less; and distribution within the continental limits of the United States.

NOTICE: When Government drawings, specifications, or other data are used for any purpose other than in connection with a definitely related Government procurement operation, the United States Government thereby incurs no responsibility nor any obligation whatsoever; and the fact that the Government may have formulated, furnished, or in any way supplied the said drawings, specifications, or other data is not to be regarded by implication or otherwise as in any manner licensing the holder or any other person or corporation, or conveying any rights or permission to manufacture, use or sell any patented invention that may in any way be related thereto.

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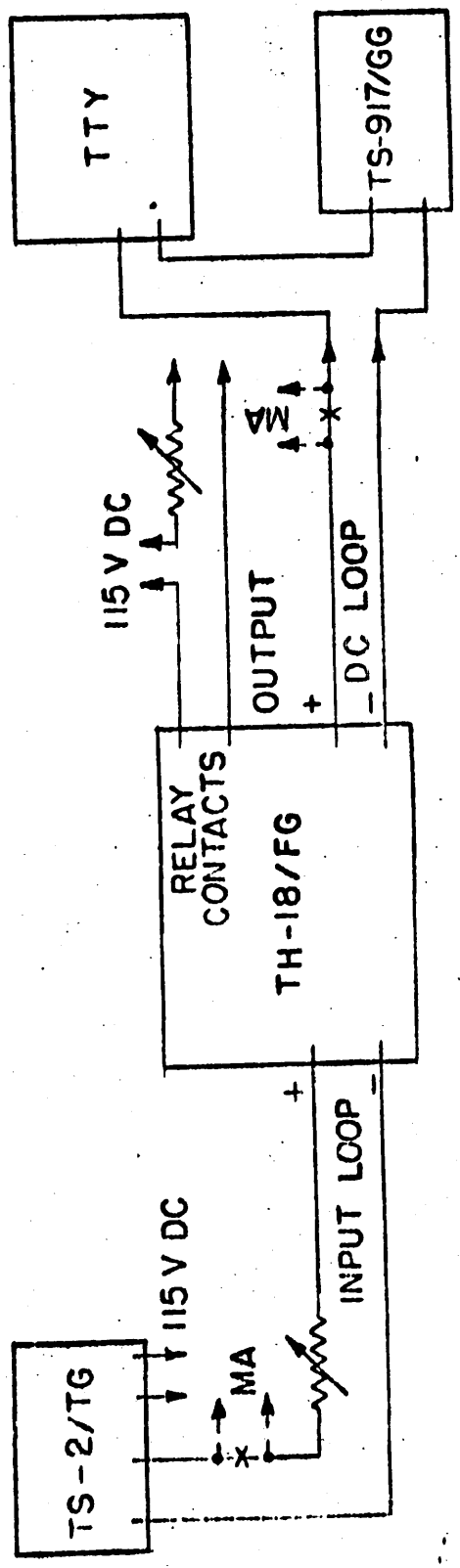


FIGURE 1

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